1 The ratio of dogs to cats is 5:3
The ratio of fish to dogs is 6:1

Find the ratio of cats to fish.
Give your answer in its simplest form.

D : C
5 : 3

F : D
6 : 1

× 5

Make Dogs the same.

D : C
5 : 3

F : D
30 : 5

D : C : F
5 : 3 : 30

C : F
3 : 30
1 : 10

1 : 10

(Total for question 1 is 2 marks)

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2 Given that $a:b = 4:5$ and $b:c = 3:2$

Find the ratio $a:b:c$
Give your answer in its simplest form.

a : b
4 : 5

b : c
3 : 2

× 3

× 5 (MAKE b the SAME)

12 : 15

15 : 10

a : b : c
12 : 15 : 10

12 : 15 : 10

(Total for question 2 is 2 marks)

3 Alfie, Bertie and Charlie share £66.
The amount Alfie and Bertie get is in the ratio 9:5
The amount Bertie and Charlie get is in the ratio 2:1

How much does Alfie get?

A : B
9 : 5
x2
18 : 10

B : C
2 : 1
x5 (MAKE B THE SAME)
10 : 5

A : B : C
18 : 10 : 5 (33 PARTS)

$\frac{66}{33}$ = £2 (PER PART)

A : B : C
£36 : £20 : £10

£ 36

(Total for question 3 is 3 marks)

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4 Dylan, Eli and Fabian share some sweets.
The amount of sweets Dylan gets to the amount of sweets Eli gets is in the ratio 7:3
The amount Dylan gets to the amount Fabian gets is in the ratio 4:5

Given Fabian gets 21 more sweets than Dylan.

Work out how many sweets Eli gets.

D : E
7 : 3
x4
28 : 12

D : F
4 : 5
x7 MAKE D THE SAME
28 : 35

D : E : F
28 : 12 : 35

FABIAN GETS 7 MORE PARTS THAN DYLAN
7 PARTS = 21 SWEETS
1 PART = 3 SWEETS

ELI GETS 12 x 3 = 36 36 sweets

(Total for question 4 is 3 marks)

5 Given that $a:b = 3:7$ and $a:c = 4:3$

Find the ratio $a:b:c$
Give your answer in its simplest form.

a : b
3 : 7
×4
12 : 28

a : c
4 : 3
×3 (Make A The Same)
12 : 9

a : b : c
12 : 28 : 9

12 : 28 : 9

(Total for question 5 is 2 marks)

6 Given that $a:c = 1:6$ and $b:c = 2:5$

Find the ratio $a:b:c$
Give your answer in its simplest form.

a : c
1 : 6
×5
5 : 30

b : c
2 : 5
×6 Make C the Same
12 : 30

a : b : c
5 : 12 : 30

5 : 12 : 30

(Total for question 6 is 2 marks)

7 There are red sweets, blue sweets and green sweets in a bag.
The ratio of red sweets to sweets that are not red is 2:3
The ratio of green sweets to sweets that are not green is 6:19

Work out the ratio of red sweets to blue sweets to green sweets.

Red to Not Red

2 : 3

5 PARTS

× 5

10 : 15

Green to Not Green

6 : 19

25 PARTS

(MAKE THE NUMBER OF PARTS THE SAME.)

R : B : G

10 : 9 : 6

$\boxed{25 - 10 - 6 = 9}$
BLUE

10 : 9 : 6

(Total for question 7 is 3 marks)

8 A football team plays some games in a season. Each game was a win, a draw or a loss.

The ratio of the games the won to the games they did not win was 9:7
The ratio of games they lost to games they did not lose was 1:7.

Given the team played less than 50 games, work out the highest amount of games they could have won.

win to not win

9 : 7

(16 PARTS)

lose to not lose

1 : 7

(8 PARTS)

× 2

2 : 14

W : D : L

(16) 9 : 5 : 2

(32) 18 : 10 : 4

(48) 27 : 15 : 6

$\boxed{\text{DRAWS } 16 - 9 - 2 = 5}$

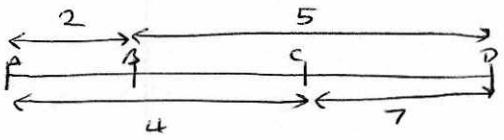
27

(Total for question 8 is 2 marks)

9 The points A, B, C and D lie in order on a straight line.

$AB:BD = 2:5$ and $AC:CD = 4:7$

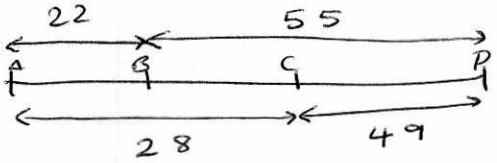
Find $AB:BC:CD$



7 PARTS × 11

11 PARTS × 7

MAKE THE NUMBER OF PARTS THE SAME



77 PARTS

77 PARTS

AB : BC : CD

22 : 6 : 49

BC = 77 − 49 − 22
= 6

22 : 6 : 49

(Total for question 9 is 3 marks)

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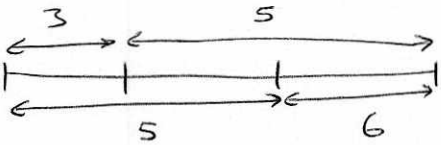
10 The points A, B, C and D lie in order on a straight line.

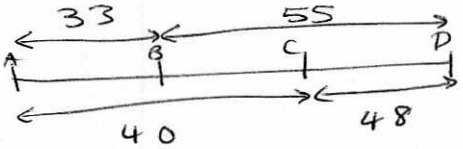
$AB:BD = 3:5$ and $AC:CD = 5:6$

Find $AB:BC:CD$



8 PARTS × 11

11 PARTS × 8



AB : BC : CD

33 : 7 : 48

33 : 7 : 48

(Total for question 10 is 3 marks)

11 Andy and Bruce share some sweets in the ratio 9:4.
Andy gets A sweets
Bruce gets B sweets

Carla and David share the same amount of sweets as Andy and Bruce.
They share their sweets in the ratio 5:2.

Carla gets C sweets
David gets D sweets

Find $A:B:C:D$

A : B
9 : 4
13 PARTS

C : D
5 : 2
7 PARTS

MAKE THE NUMBER OF PARTS THE SAME

x 7
63 : 28

x 13
65 : 26

A : B : C : D
63 : 28 : 65 : 26

63 : 28 : 65 : 26

(Total for question 11 is 3 marks)

12 A and B are in the ratio 5:1
C and D are in the ratio 2:3

Given: $A + B = 2(C + D)$

Find $A:B:C:D$

A : B
5 : 1
6 PARTS
x 5
25 : 5

C : D
2 : 3
5 PARTS
x 6
12 : 18
÷ 2
6 : 9

$A + B = 2(C + D)$

A : B : C : D
25 : 5 : 6 : 9

(Total for question 12 is 3 marks)

13 Glen and Harper share some money in the ratio 5:2.
Glen gets £*G*
Harper gets £*H*

India and Jade share the same amount of money as Glen and Harper.
They share their money in the ratio 4:7.

India gets £*I*
Jade gets £*J*

Find *G*:*H*:*I*:*J*

G : H I : J
5 : 2 (7) 4 : 7 (11)
×11 ×7
55 : 22 28 : 49

G : H : I : J
55 : 22 : 28 : 49

55 : 22 : 28 : 49

(Total for question 13 is 3 marks)

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14 *A* and *B* are in the ratio 4:3
C and *D* are in the ratio 1:5

Given: $3(A + B) = C + D$

Find *A*:*B*:*C*:*D*

A : B C : D
4 : 3 (7) 1 : 5 (6)
×6 ×7
24 : 18 7 : 35
÷ 3
3 (A+B) = C+D
8 : 6

A : B : C : D
8 : 6 : 7 : 35

8 : 6 : 7 : 35

(Total for question 14 is 3 marks)